

Sample Formulations for the LTQ XL and LXQ Mass Spectrometers

This appendix describes how to prepare the sample and calibration solutions for the LTQ XL and LXQ MSs. You can order a replacement Chemical Kit, OPTON-20041 (P/N 97355-62070), from Thermo Fisher Scientific. You can also order specific chemicals from Thermo Fisher Scientific, which are sold under its Fisher Chemical brand. For a complete selection of LC/MS-grade consumables, visit www.fishersci.com.

IMPORTANT NEVER use the solutions described in this appendix to calibrate the Velos Pro MS. Instead, use the solutions described in [Appendix A, “Sample Formulations for the Velos Pro Mass Spectrometer.”](#)

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CAUTION Avoid exposure to potentially harmful materials.

By law, producers and suppliers of chemical compounds are required to provide their customers with the most current health and safety information in the form of Material Safety Data Sheets (MSDSs) or Safety Data Sheets (SDSs). The MSDSs and SDSs must be freely available to lab personnel to examine at any time. These data sheets describe the chemicals and summarize information on the hazard and toxicity of specific chemical compounds. They also provide information on the proper handling of compounds, first aid for accidental exposure, and procedures to remedy spills or leaks.

Read the MSDS or SDS for each chemical you use. Store and handle all chemicals in accordance with standard safety procedures. Always wear protective gloves and safety glasses when you use solvents or corrosives. Also, contain waste streams, use proper ventilation, and dispose of all laboratory reagents according to the directions in the MSDS or SDS.

Use only LC/MS-grade chemicals for calibrating the instruments. For suggested consumable part numbers, refer to Chapter 5, “Gases and Solvents,” in the *LTQ Series Preinstallation Requirements Guide*.

Potentially hazardous chemicals used in the procedures throughout this appendix include the following:

- Glacial acetic acid
- Methanol
- Reserpine

IMPORTANT

- Do not filter solvents. Filtering solvents can introduce contamination
- Do not use plastic pipettes to prepare the tuning and calibration solutions. Plastic products can release phthalates that can interfere with the analyses.

Preparing the Reserpine Sample Solution for the ESI and APCI Modes

Ideally, you should prepare the reserpine solution just before using it. If you must store the solution, keep it in a light-resistant container in the refrigerator until needed.

❖ To prepare the reserpine sample solution

1. Transfer 100 μL of the 100 $\text{pg}/\mu\text{L}$ reserpine stock solution into a clean, minimum 1.5 mL polypropylene tube.
2. Add 900 μL of 1% acetic acid in 50:50 methanol/water to the tube.
3. Mix the solution (10 $\text{pg}/\mu\text{L}$) thoroughly.
4. Transfer 100 μL of the 10 $\text{pg}/\mu\text{L}$ solution into a clean, minimum 1.5 mL polypropylene tube.
5. Add 900 μL of 1% acetic acid in 50:50 methanol/water to the tube.
6. Mix the solution (1 $\text{pg}/\mu\text{L}$) thoroughly.
7. Transfer 100 μL of the 1 $\text{pg}/\mu\text{L}$ solution into a clean, minimum 1.5 mL polypropylene tube.
8. Add 700 μL of 1% acetic acid in 50:50 methanol/water to the tube.
9. Mix the solution thoroughly.
10. Label the tube **Reserpine Sample Solution (125 $\text{fg}/\mu\text{L}$)**.

Normal Mass Range Calibration Solution for ESI Mode

For tuning and calibrating the LTQ XL and LXQ in ESI mode, use a calibration solution of caffeine, MRFA, and Ultramark 1621 in an acetonitrile/methanol/water solution containing 1% acetic acid.

Tip You can order the following ready-to-use normal mass range calibration solution from Thermo Fisher Scientific at www.thermoscientific.com/pierce:

Pierce LTQ ESI Positive Ion Calibration Solution, 10 mL (Pierce P/N 88322)

Preparing the High Mass Range Calibration Solution

The high mass range calibrant is a solution of 3.5 mg/μL polypropylene glycol (PPG) in a solvent of 70:30 methanol/23 mM sodium acetate. The high mass range calibration procedure is designed to work with a PPG that has an average molecular weight of approximately 2700 ($M_n \sim 2700$), which is Sigma-Aldrich product number 202347. PPG 2700 is a viscous liquid. To order this compound from Sigma-Aldrich, write or call:

Sigma Chemical Company
P. O. Box 14508, St. Louis, Missouri U.S. 63178-9916
(U.S.) 1 (800) 325-3010; (Canada) 1 (905) 829-9500
(Outside the U.S. or Canada) (314) 771-3750
www.sigmaaldrich.com

To prepare the high mass range calibration solution, follow these procedures:

1. [To prepare the sodium acetate stock solution](#)
2. [To prepare the PPG stock solution](#)
3. [To prepare the calibration solution](#)

❖ To prepare the sodium acetate stock solution

Dissolve 0.082 g of sodium acetate in 10 mL of water in a clean 20 mL glass vial and label the container **Sodium Acetate Stock Solution**.

❖ To prepare the PPG stock solution

1. Dissolve 0.7 g of the PPG 2700 in 7 mL of methanol in a clean 20 mL glass vial.

Tip Because PPG 2700 is a viscous liquid, use a glass pipette to transfer 0.7 g of the liquid into a weigh boat, or weigh the liquid directly into a minimum 20 mL glass vial.

2. Add 2.3 mL of water to the vial.
3. Add 0.7 mL of the sodium acetate stock solution and label the container
PPG 2700 Stock Solution (70 mg/mL).

❖ To prepare the calibration solution

1. Pipette 7 mL of methanol into a clean 20 mL glass vial.
2. Add 2.3 mL of water to the vial.
3. Add 700 μ L of the sodium acetate stock solution to the vial.
4. Add 10 μ L of the PPG 2700 stock solution and label the container
LTQ XL/LXQ PPG 2700 Calibration Solution (70 ng/ μ L).